ASCII MASTER FLOW — PANEL 1/12: Nuclear-process firewall

FISSION -> splits heavy nuclei and powers current reactors
FUSION -> combines light nuclei in experimental systems
COMMERCIAL STATUS -> fission operating, fusion not deployed
SHARED WORD NUCLEAR -> does not erase process differences
RULE -> Topic 05 cannot substitute for Topic 04

ASCII MASTER FLOW — PANEL 2/12: D-T energy map

DEUTERIUM + TRITIUM -> fusion reaction
HELIUM-4 / ALPHA -> charged product retained for self-heating
NEUTRON -> carries most energy out of plasma
BLANKET -> captures heat and supports fuel functions
RULE -> particle role controls engineering consequence

ASCII MASTER FLOW — PANEL 3/12: Fusion-condition triangle

TEMPERATURE -> raises probability of overcoming repulsion
DENSITY -> supplies reacting particles
CONFINEMENT TIME -> retains energy long enough
LAWSON TRIPLE PRODUCT -> all three must combine
TRAP -> a temperature record alone proves little

ASCII MASTER FLOW — PANEL 4/12: Confinement comparison

MAGNETIC -> lower density and longer confinement
TOKAMAK -> torus with strong plasma current
STELLARATOR -> externally shaped steady-field geometry
INERTIAL -> dense target and extremely short confinement
RULE -> approach and device name are not power status

ASCII MASTER FLOW — PANEL 5/12: Breakeven ladder

PLASMA Q -> fusion power divided by injected heating
Q ABOVE ONE -> plasma-level scientific threshold
ENGINEERING Q -> includes recirculating plant loads
NET ELECTRICITY -> includes thermal conversion and all consumption
COMMERCIALITY -> adds availability, cost and licensing

ASCII MASTER FLOW — PANEL 6/12: Neutron-to-material chain

```
14.1 MeV NEUTRON -> escapes magnetic confinement
BLANKET -> receives energy and may breed tritium
STRUCTURE -> displacement damage and transmutation
ACTIVATION -> radioactive material-management issue
LIFETIME -> separate engineering feasibility constraint
```

ASCII MASTER FLOW — PANEL 7/12: Tritium fuel loop

```
D-T PLASMA -> consumes tritium
FUSION NEUTRON -> enters lithium-bearing blanket
LITHIUM REACTION -> produces new tritium
EXTRACTION / PROCESSING -> returns usable fuel
RULE -> a test blanket is not a closed commercial cycle
```

ASCII MASTER FLOW — PANEL 8/12: Plasma exhaust path

```
CORE PLASMA -> creates heat and helium ash
EDGE TRANSPORT -> moves particles and heat outward
DIVERTOR -> exhausts impurities and helium
PLASMA-FACING MATERIAL -> receives extreme heat load
RULE -> confinement success does not solve exhaust lifetime
```

ASCII MASTER FLOW — PANEL 9/12: ITER mission card

```
UNDER CONSTRUCTION -> experimental international tokamak
BURNING-PLASMA STUDY -> core scientific purpose
Q=10 DESIGN -> 500 MW fusion from 50 MW plasma heating
NO TURBINE -> no conversion to grid electricity
TEST TECHNOLOGIES -> bridge to later machines
```

ASCII MASTER FLOW — PANEL 10/12: India-in-ITER map

```
INDIA -> one of seven ITER members
IN-KIND PACKAGES -> specified hardware and systems
IPR -> DAE-aided plasma-research institute
ITER-INDIA -> domestic procurement and delivery agency in IPR
RULE -> membership, institute, agency and package differ
MUST REMEMBER: Fusion joins light nuclei in a high-temperature plasma; confinement,...
```

ASCII MASTER FLOW — PANEL 11/12: Domestic research devices

```
ADITYA-U -> experimental tokamak and plasma campaigns
SST-1 -> superconducting long-pulse research platform
RESEARCH OUTPUT -> diagnostics, control and operating knowledge
NOT OUTPUT -> commercial fusion electricity
RULE -> experimental operation is not power generation
CLOSE DISTINCTION: Plasma Q is not engineering breakeven, tokamak is not stellarator,...
```

ASCII MASTER FLOW — PANEL 12/12: Fusion-readiness answer spine

```
DEFINE -> D-T fusion plasma confinement Lawson and Q
TRACE -> alpha heating neutron blanket tritium and materials
SEPARATE -> plasma gain engineering gain and net electricity
MAP -> ITER India IPR ITER-India ADITYA-U SST-1 and DEMO
VERIFY -> dated schedule cost contribution and milestone evidence
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State quantity, device, pulse/energy...
```